

INVESTIGATION REPORT

Issue 1012: Debug flush with an impossible single cause

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1012
Disposition	Dead code, proven
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1012
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL master 6033d2b, COREV_PULP=1 FPU=0
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	6 cited, 5 present, 1 not retained

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The three uncovered rows (4, 5, 10) of the if-condition at cv32e40p_controller.sv:1241 (state DBG_FLUSH, no data error) are unreachable. Row 5 (ebreak instruction present but ebrk_force_debug_mode clear, all other causes clear) and row 10 (data_load_event_i as the only cause) are proven by unbounded abc pdr in 131.5 s and 126.8 s. Row 4 (trigger_match_i as the only cause) resists pdr because its cone includes the 32-bit trigger comparator; it is proven instead by k-induction (smtbmc/yices, k=12) in 45.0 s: base case and inductive step both pass, which is a full unbounded proof. Mechanism: every entry into DBG_FLUSH caused by a trigger or debug request sets debug_req_entry_q in the same cycle (controller lines 491-498, 723-729), so trigger_match_i can never be the sole true term; the only DBG_FLUSH entry that leaves debug_req_entry_q clear is single-step, taken only when trigger_match_i was already 0, and the ID stage is halted so pc_id and the tmatch CSRs cannot change while in DBG_FLUSH. Anti-vacuity: bounded cover reaches the DBG_FLUSH branch and the taken-branch at step 7. Environment assumptions: OBI handshake legality, scan_cg_en_i=0, fetch_enable_i=1. Configuration COREV_PULP=1, FPU=0.

3 Conditions

- 1. OBI handshake legality
- 2. scan_cg_en_i=0
- 3. fetch_enable_i=1

Extracted verbatim from the environment assumptions stated in the verdict record for this run.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:1241 rows 4/5/10 (DBG_FLUSH cause selection): formally proven unreachable with open-source tools. Rows 5 and 10: abc pdr unbounded proof. Row 4: k-induction (k=12) unbounded proof. Any trigger/debug-request entry into DBG_FLUSH sets debug_req_entry_q in the same cycle, so trigger_match_i, ebreak-without-force, or data_load_event_i can never be the sole cause. Evidence: runs/v1-cv32e40p-1012/formal/{prove_row5,prove_row10,kind_row4,cover}. Exclude the three rows from condition coverage.

Checker used

checker_1012.sv, checker_1012_row4.sv, checker_1012_row5.sv, checker_1012_row10.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/prove_row5/sby.log	present	abc pdr PASS (131.5 s): row 5 unreachable
E2	proof	formal/prove_row10/sby.log	present	abc pdr PASS (126.8 s): row 10 unreachable
E3	proof	formal/kind_row4/sby.log	present	k-induction (smtbmc yices, k=12) PASS basecase + induction (45.0 s): row 4 unreachable
E4	proof	formal/cover/sby.log	present	cover PASS: DBG_FLUSH branch and taken-branch reached at step 7 (covers_reached=2) -> not vacuous
E5	counterexample	formal/cover/job/engine_0/trace0.vcd	not kept	witness trace of DBG_FLUSH entry, rendered in cover_dbg_flush.png
E6	proof	formal/prove/sby.log	present	negative result kept for honesty: combined 3-assert pdr TIMEOUT 30 min (rows 5,10 proved inside it, row 4 not)

Paths are relative to runs/v1-cv32e40p-1012. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E5 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory runs/v1-cv32e40p-1012

Title source Title and configuration read back from reports/issues/cv32e40p_issue_1012.pdf, the

Inputs shown	previously published report for this issue.
Human record	title, body(links stripped) Dolphin Design 2024: waived; Questa/OneSpin asserts assert_unreachable_ctrl_1241_row_4/5/10 in scripts/formal
Agent confidence	0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	4
Run by	Claude directly (no headless agent)

Measured tool and step time

Step	Measured
fetch + locate + classify (manual)	5.0 min (300 s)
combined pdr attempt (timeout)	30.0 min (1800 s)
split: row5 pdr	2.2 min (132 s)
split: row10 pdr	2.1 min (127 s)
row4 pdr + lemma attempts (timeout, parallel)	30.0 min (1800 s)
row4 k-induction	45.0 s
cover	33.0 s
report	10.0 min (600 s)
Total	66.0 min (3960 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Note. wall 16:09 -> 17:15 including 4 timed-out pdr attempts; successful jobs alone 5.1 min

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